



CJC

Complete Java Classes

by Kunal Sir

Data Types/ Data Member in Java: -

What are Data Types?

Data types mean to identify the type of data and associate operations that can be done on the data values.

Data types define the values that a variable can take.

Java is **Statically Typed** Programming Language.

What is Statically Typed Programming language?

Statically typed languages are those languages in which the type of data values or expression that variable will hold is decided during the compilation time of the program is called statically typed language.

Is Java Statically Typed Programming language?

Yes, Java language is a statically typed language, which means the type of particular variable is defined during the compilation time of the program.



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```
Scanner scan=new Scanner(System.in);  
  
int number=scan.nextInt(); //number variable will store  
                           integer value  
  
number="3242" //Compiler will show error
```

That means in the above snippet the number variable is defined for storing int values if we try to assign other data values the java compiler will raise an error that is why java language is statically typed language.

Types of Data Types in Java: -

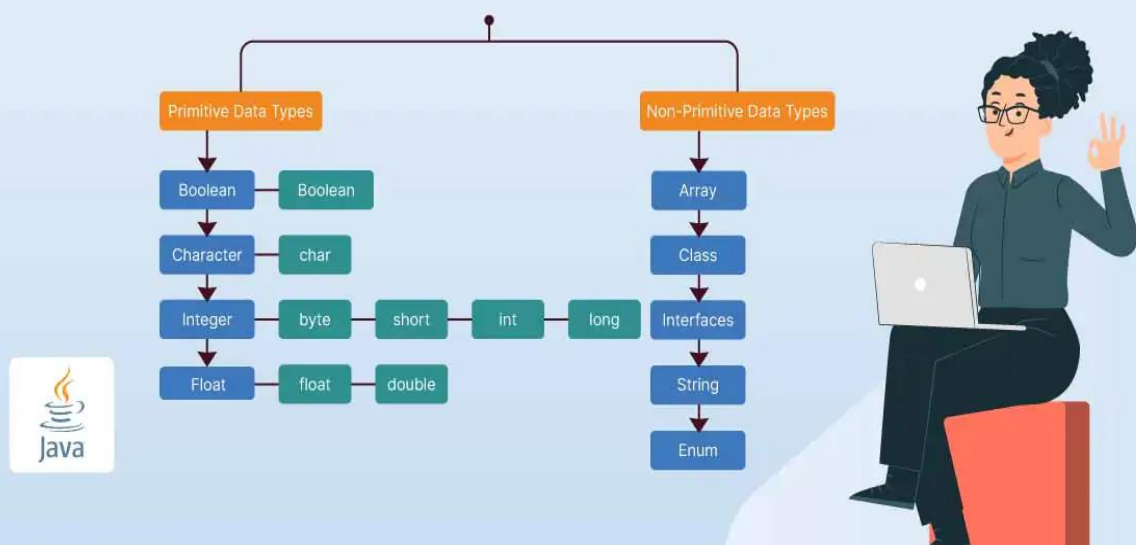
There are two types of data types

1. Primitive Data Type

2. Non-Primitive Data Type



Data Types in Java



1. Primitive Data Type: -

Primitive data types are those data types that are predefined in the java programming language.

There are a total of **eight primitive data types** that are predefined in the java programming language.

The size of primitive data types is fix.

Now we can see the default value, size and range of these data types are given below-



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Data Types	Size	Default Value	Range
int	4bytes	0	-2^{31} to $2^{31}-1$
float	4bytes	0.0	6-7 decimal dig
boolean	1bit	false	true/false
char	2bytes	blank space	0 to 65,535
byte	1bytes	0	-128 to 127
short	2bytes	0	-32768 to 32767
double	8bytes	0.0	15 decimal dig.
long	8bytes	0	-2^{63} to $2^{63}-1$

2. Non-Primitive Data Type: -

These are the data types that are not predefined but defined by the programmer according to the need for a particular task.

Default value of non-primitive data type is **"null"**.



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Everything inside the java except the primitive data type is non primitive data types

For example – String, Arrays, Class etc.

Non- primitive is also known as user-define data type.

Non- primitive is also known as reference data type.



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